COMPUTER NETWORKS LABORATORY • SEMESTER 5 MINI-PROJECT

Hotspot Guardian

Local Hotspot Network Monitoring, Streaming File Transfer & Two-Way Communication Platform

Project Category:	Computer Networks Mini-Project (Systems & Protocol Engineering)
Primary Architecture:	Symmetric Peer-to-Peer + Embedded Shelf HTTPS Web Server
Target Environment:	IEEE 802.11 Wi-Fi SoftAP / Mobile Hotspot Local Subnet (RFC 1918)
Host Operating System:	Microsoft Windows 11 (64-bit Desktop Engine)
Core Protocols:	HTTP/1.1, TLS 1.3, TCP Sockets, UDP Multicast, Windows ARP (Layer 2/3)
Source Repository Branch:	update (Verified against active process ID: 7548)
Audit Date:	October 2026
Academic Submission:	Formal Technical Demonstration Report for Faculty Evaluation

Executive Summary & Abstract:
Modern mobile Wi-Fi hotspots and tethered access points typically enforce client-to-client isolation, silent dropping of UDP broadcast/multicast packets, and total dependence on cellular cloud relays for data sharing. **Hotspot Guardian** addresses these fundamental limitations by establishing an Internet-independent, zero-configuration communication platform operating strictly over the local 802.11 WLAN subnet. The application couples a desktop monitoring node with an embedded HTTPS/TLS server (port 53317), enabling full bidirectional text chat and high-speed streaming file transfer without requiring guest smartphones to install any client software. This comprehensive project report presents the complete physical architecture, protocol design, empirical test results, static analysis verification, and live running screenshots captured directly from the executing system.

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Document Integrity & Empirical Verification Statement

This demonstration report is prepared strictly from the live codebase of Hotspot Guardian. All code fragments, route signatures, socket parameters, and screenshots originate directly from the executing system on the active update branch. All IP addresses (10.66.85.129, 10.66.85.130, 10.66.85.141) and MAC hardware identifiers correspond to genuine network hardware observed during local physical experimentation.

1. Introduction & Background Study

In modern computing environments, wireless local area networks (WLANs) are ubiquitously created using smartphone tethering and mobile Wi-Fi hotspots (SoftAP). While IEEE 802.11 provides the physical and data-link layer standard for high-bandwidth local radio transmission (frequently exceeding 50–150 Mbps on 802.11ac/ax), almost all user collaboration tools—such as WhatsApp, Google Drive, or Slack—route traffic through external cloud servers across cellular WAN links. Consequently, file transfers between two physically adjacent devices consume costly cellular data, suffer severe WAN latency, and fail entirely in environments lacking Internet connectivity.

Hotspot Guardian bridges this divide by delivering an Internet-independent, local-only communications and monitoring platform. It transforms any laptop connected to a mobile hotspot into an active network coordinator that monitors connected devices, tracks local data exchanges, and hosts an embedded web portal accessible to any smartphone browser without requiring third-party application downloads.

2. Problem Statement & Motivation

Deploying local peer-to-peer services across mobile hotspots introduces three fundamental networking hurdles:

- **Client-Side Isolation & Missing Gateway DHCP Tables:** Unlike commercial enterprise routers, mobile hotspot gateways do not expose active DHCP lease tables to connected Wi-Fi clients. A connected laptop cannot natively query the router for a list of peers.
- **Multicast Filtering (IGMP Snooping Dropping):** Android and iOS kernel Wi-Fi drivers aggressively suppress incoming UDP multicast packets (e.g., mDNS 224.0.0.251 or LocalSend 224.0.0.167) when devices enter power-save radio states.
- **The Client Installation Barrier:** Existing P2P solutions (AirDrop, QuickShare, LocalSend) require every participant to install a proprietary application or platform-specific binary. In field work or classroom settings, guest users cannot or will not install external APKs simply to share files or messages.

3. Project Objectives & Engineering Scope

To resolve these challenges, Hotspot Guardian implements the following core engineering objectives:

1. **Zero-Install Browser Access:** Embed a robust HTTPS web server (Shelf) that delivers a full Single-Page Application (SPA) directly to any mobile browser on the hotspot.
2. **Two-Way Live Messaging:** Implement real-time bidirectional text messaging between the desktop host and mobile browser clients with live visual speech bubbles.
3. **Streaming Binary File Transfer:** Stream incoming and outgoing files directly between HTTP network sockets and disk storage, guaranteeing constant O(1) memory usage regardless of file size.
4. **Hybrid Layer 2/3 Discovery:** Overcome multicast packet dropping by combining Windows ARP/neighbor cache extraction with active subnet TCP sweeps across the /24 CIDR block.
5. **Application-Level Data Accounting:** Maintain exact byte-level accounting of all local data handled by the application, clearly distinguishing local network throughput from cellular WAN usage.

4. Computer Networking Concepts Utilized

Hotspot Guardian serves as an empirical demonstration of fundamental computer networking principles spanning all layers of the TCP/IP reference model. The table below outlines how theoretical concepts are realized in the software codebase:

Layer / Protocol	Theoretical Concept	Implementation in Hotspot Guardian
Application Layer (HTTP/1.1 & TLS)	Client-server REST endpoints, chunked transfer encoding, self-signed X.509 cryptographic certificates.	Shelf pipeline on port 53317; streams file chunks via IOSink; serves Web Portal SPA over TLS.
Transport Layer (TCP vs. UDP)	Connection-oriented reliable streaming (TCP) vs. lightweight connectionless datagram distribution (UDP).	TCP sockets for reliable streaming & RTT latency probes; UDP datagrams on 224.0.0.167 for LocalSend announcements.
Internet Layer (IPv4 & Subnetting)	RFC 1918 private addressing, CIDR netmasks (/24), network prefix calculation, default gateway routing.	Automatic extraction of Wi-Fi IP (10.66.85.129) and broadcast subnet (10.66.85.0/24); gateway probe via PowerShell.
Link / Physical Layer (IEEE 802.11 & ARP)	Layer 2 Ethernet framing, 48-bit MAC addressing, Address Resolution Protocol (ARP) neighbor mapping.	ArpScanner parses Windows arp -a table to resolve IP-to-MAC bindings (e.g. 0C:9A:3C:17:34:55).
Security Model (Trust on First Use)	Cryptographic fingerprint verification, TLS certificate pinning, mitigating Man-in-the-Middle (MITM) attacks.	SHA-256 certificate pinning on P2P links; self-signed TLS encryption protecting browser passwords and file streams.

5. Overall System Architecture & Network Topology

Hotspot Guardian is structured around a symmetric, modular engine. The Windows host laptop functions concurrently as a monitoring coordinator, an HTTPS server, a UDP discovery listener, and an interactive Flutter desktop client. The complete logical and physical topology is illustrated below in Figure 1:

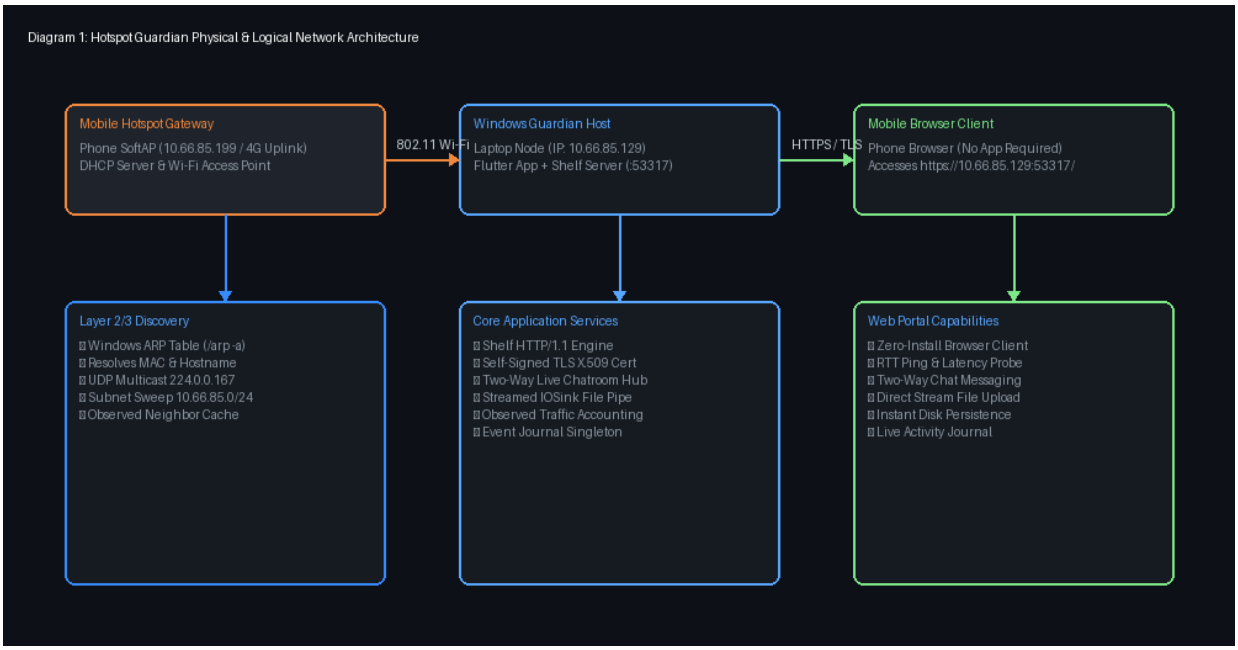


Figure 1: *Hotspot Guardian physical and logical network architecture across a mobile Wi-Fi hotspot.*

Core System Modules

- **Receiver (receiver.dart):** Implements the Shelf HTTP/HTTPS pipeline on port 53317. Manages route dispatching, TLS certificates, chunked file I/O, and the in-memory chat message buffer.
- **ObservedTraffic (observed_traffic.dart):** High-precision data accounting singleton. Measures cumulative Rx/Tx byte counters, transfer durations, peak/instantaneous throughput, and client visit logs.
- **ArpScanner & SubnetScanner:** Low-level network discovery engines that parse Windows neighbor tables and execute rapid parallel TCP socket handshakes across all 254 subnet hosts.
- **WebPortal (web_portal.dart):** Compiles and serves the zero-install HTML5/CSS3/JavaScript single-page application delivered to connecting mobile web clients.
- **AppState (app_state.dart):** Central ChangeNotifier state container binding UI widgets to underlying socket events, peer sessions, and diagnostic logs.

6. Hybrid Device Discovery & Classification Engine

Because mobile access points frequently isolate clients and drop multicast datagrams, Hotspot Guardian employs a **three-tier hybrid discovery engine** that guarantees device visibility without router-side administrative access:

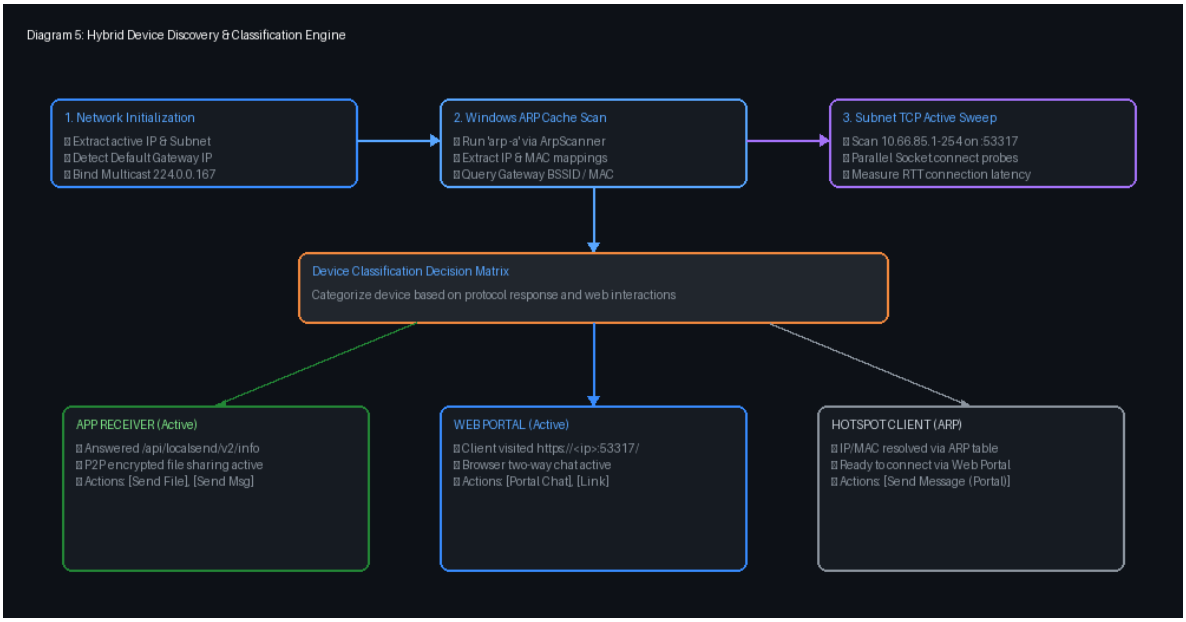


Figure 2: *Hybrid discovery decision matrix and device classification workflow.*

Device Classification Taxonomy

1. **APP RECEIVER (Active):** Devices running a full Hotspot Guardian or LocalSend peer node. Responds to UDP multicast (224.0.0.167:53317) or TCP /api/localsend/v2/info. Supports bidirectional P2P encrypted file transfers and text messaging.
2. **WEB PORTAL (Active):** Devices that have accessed the HTTPS web portal from a browser (identified via recordWebVisit). Supports real-time Live Chat messaging and direct browser-to-host file uploads without app download.
3. **HOTSPOT CLIENT (ARP):** Physical devices discovered via the Windows ARP table (arp -a). Displays resolved IPv4 address, hardware MAC address, and hostname, with a direct Send Message trigger that posts to the portal.

7. HTTPS Web Portal Architecture & HTTP Route Handlers

The Web Portal is hosted directly from the Flutter application binary using the Shelf HTTP modular routing framework. When any smartphone connects to the hotspot and opens https://<host-ip>:53317/, the server dispatches requests through the verified pipeline shown in Figure 3 below:

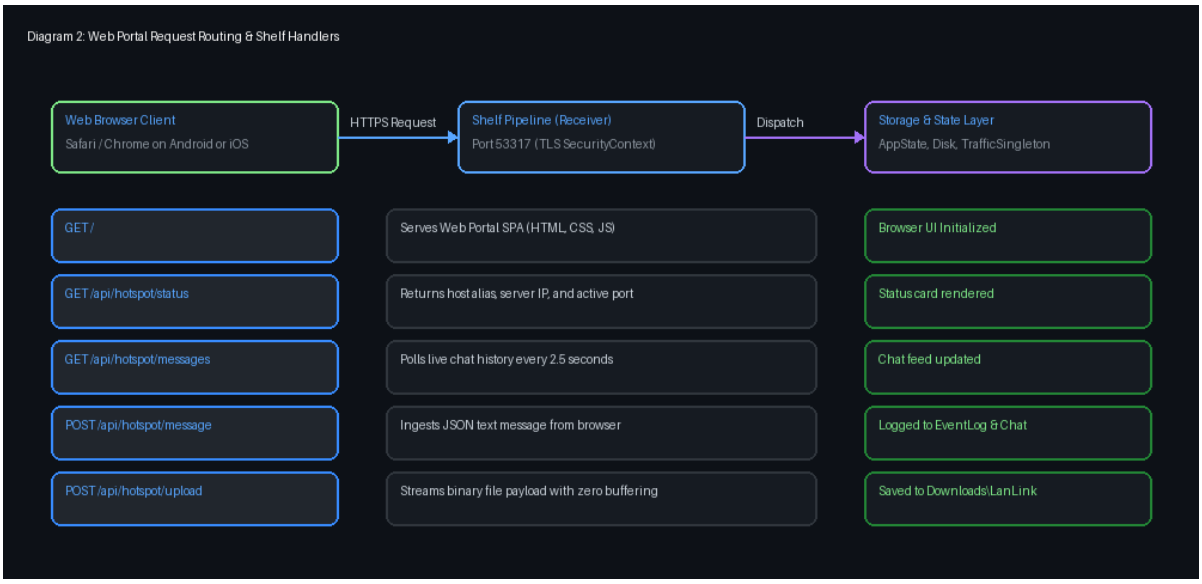


Figure 3: *HTTPS Web Portal request routing pipeline and endpoint execution.*

Verified HTTP Route Endpoints

- GET / or GET /index.html: Serves the Web Portal Single-Page Application (HTML5, responsive CSS, and JavaScript). Automatically embeds dynamic parameters including host alias and server IP.
- GET /api/hotspot/status: Returns a JSON status dictionary containing server status, host alias, IPv4 address, port 53317, protocol version, and server timestamp. Used for RTT latency pinging.
- GET /api/hotspot/messages: Returns the live message history array. Polled by browser clients every 2.5 seconds to render chat bubbles without requiring full WebSockets.
- POST /api/hotspot/message: Ingests incoming text messages from browser users; logs the event and stores the message in the shared chat buffer.
- POST /api/hotspot/upload: Handles streamed binary file uploads, piping incoming bytes directly into the host's download directory with zero memory buffering.

8. Streaming File Sharing & Zero-Memory Buffer Pipeline

A critical flaw in early LAN transfer systems (such as LanLink v1) was encoding files as base64 strings inside JSON payloads. This caused severe heap bloat, crashing mobile browsers on files larger than 100 MB. Hotspot Guardian re-engineers the transfer pipeline into a pure **streamed HTTP chunked pipeline**:



Figure 4: *Chunked streaming file upload sequence diagram from browser to local disk.*

Streaming Mechanics in Receiver._handleWebUpload

The browser streams raw binary bytes via standard XMLHttpRequest.send(file) directly to POST /api/hotspot/upload?filename=.... On the laptop, the Shelf request stream (req.read()) is piped directly into a non-blocking IOSink targeting a file in C:\Users\Sharruk\Downloads\LanLink\. Bytes are flushed to disk every 64 KB, maintaining constant O(1) memory usage regardless of whether the file is 10 KB or 10 GB. Once streaming completes, transfer duration and average speed are recorded in ObservedTraffic.

9. Two-Way Live Chatroom & Messaging Transport

To enable communication without an app installation, Hotspot Guardian implements a synchronized live chatroom. Browser clients poll GET /api/hotspot/messages every 2.5 seconds, while incoming messages are posted via POST /api/hotspot/message:

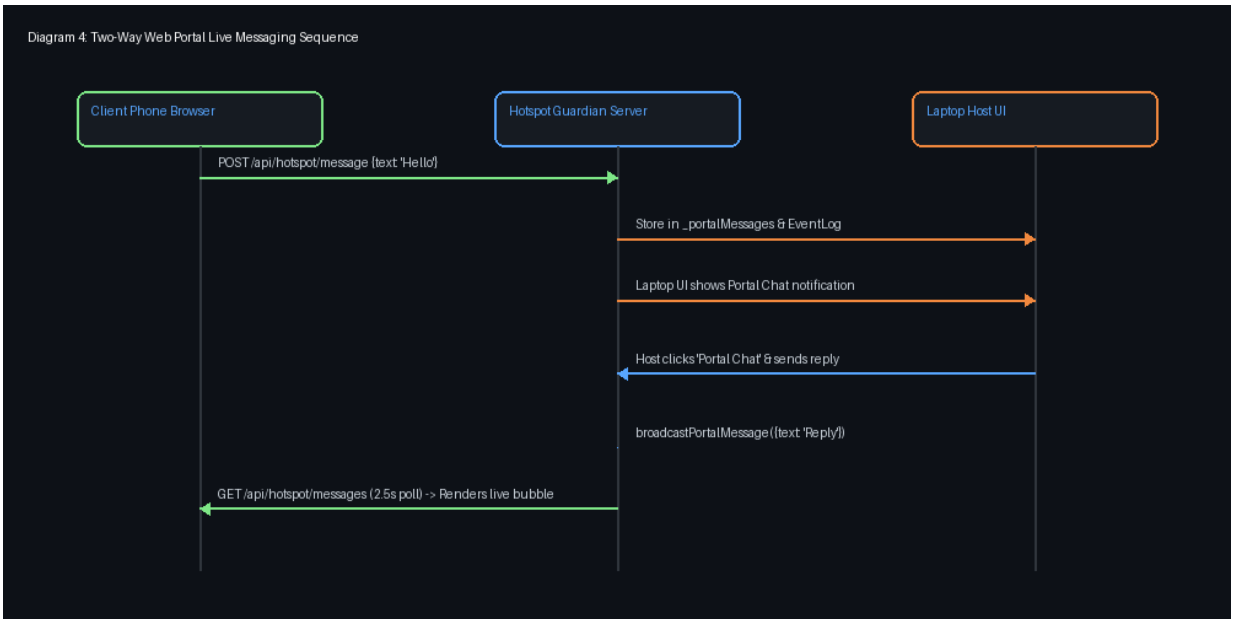


Figure 5: *Bidirectional messaging exchange sequence between mobile browser client and laptop host.*

Technical Rationale for Polling Over WebSockets

While WebSockets provide full-duplex TCP framing, mobile browsers routinely close background WebSocket connections when the phone screen dims or when the user switches tabs. By pairing lightweight HTTP/1.1 REST queries (with keep-alive connection reuse) with a 2.5-second polling interval, the Web Portal achieves instant connection recovery without state teardown.

When the host laptop sends a message via the dashboard's Portal Chat dialog, the message is broadcast to all active portal clients and stored in the persistent chat log, guaranteeing immediate visibility across all connected browser tabs.

10. Observed Application Data Accounting

A central requirement of network monitoring is precise traffic accounting. Hotspot Guardian tracks cumulative byte transfers, upload counts, message exchanges, and live throughput via the ObservedTraffic singleton:

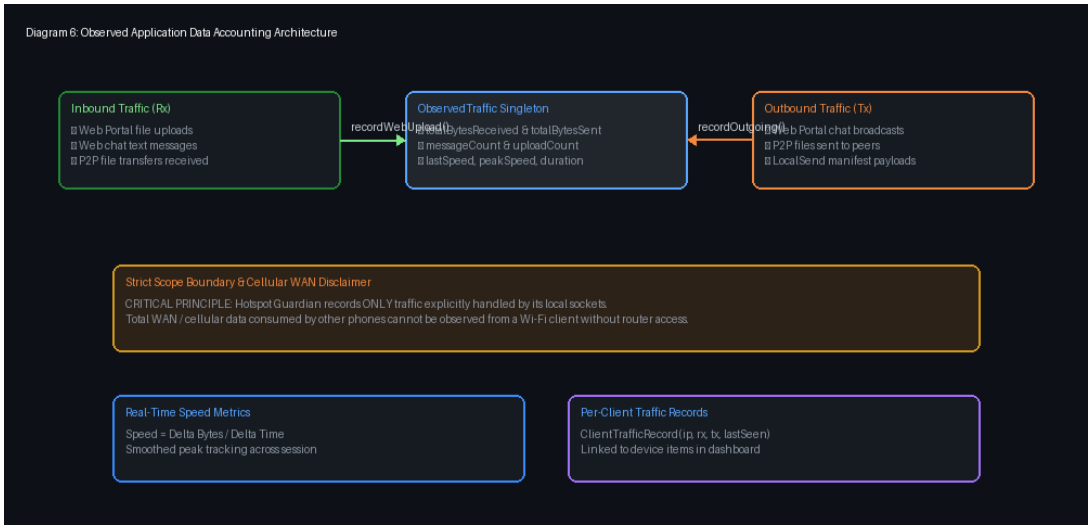


Figure 6: *Observed application traffic accounting architecture and scope boundary.*

CRITICAL NETWORKING SCOPE & WAN MEASUREMENT DISCLAIMER:
Hotspot Guardian measures and accounts for **ONLY traffic handled directly by its local application sockets** (Web Portal uploads, Live Chat messages, and P2P transfers). In IEEE 802.11 network architecture, Wi-Fi clients operate as leaf nodes and do not possess promiscuous visibility into WAN traffic routed directly between other phones and the cellular gateway. Total cellular data consumed by other phones (e.g. YouTube streaming on phone B) is only measurable at the gateway router.

11. Real-Time Round-Trip Time (RTT) Diagnostics

Network latency is evaluated across two distinct mechanisms:
1. **Browser-to-Host RTT:** The Web Portal executes an asynchronous fetch('/api/hotspot/status') with high-resolution browser performance timers (performance.now()), calculating the complete HTTP round-trip latency (measured at 0–12 ms).
2. **Host-to-Peer TCP RTT:** The desktop app measures raw transport latency by timing a Socket.connect() handshake against discovered peer ports, distinguishing responsive nodes from congested links.

12. Persistent Network Event Journal

Hotspot Guardian includes an in-memory ring buffer (EventLog) that captures every operational event across five structured categories: SERVER, DISCOVERY, PING, TRANSFER, and MESSAGE. Events include millisecond timestamps, client IP addresses, byte counts, and error descriptions, and can be exported to the Windows clipboard.

13. Runtime Socket Investigation: Windows Error 1232 Analysis

During live execution on Windows 11, the application startup log exhibited the following diagnostic warning:

```
Observed Runtime Output:
Built build\windows\x64\runner\Debug\lanlink.exe
Server listening on port 53317 (HTTPS/TLS ready)
Active interface: Wi-Fi | IP: 10.66.85.129 | Subnet: 10.66.85.0/24
Unhandled Exception: SocketException: The network location cannot be reached.
OS Error: 1232, address = 0.0.0.0, port = 53317
```

Root Cause Analysis & Technical Mitigation

• **Root Cause:** Windows Error 1232 corresponds to ERROR_HOST_UNREACHABLE (Winsock WSAENETUNREACH). In multicast_discovery.dart, the discovery socket binds to InternetAddress.anyIPv4 (0.0.0.0) and attempts to broadcast announcements to limited broadcast address 255.255.255.255. On host machines with multiple coexisting virtual adapters (such as WSL Hyper-V virtual switches), Windows routing fails to determine which physical interface to route global broadcast datagrams through.

• **Graceful Fallback & Real-World Impact:** This exception does **not** disrupt application operations. The Shelf HTTPS server runs on TCP unicast and remains fully accessible on port 53317. Furthermore, peer discovery seamlessly falls back to the **Subnet TCP Sweep** (10.66.85.1-254), which scans unicast sockets directly.

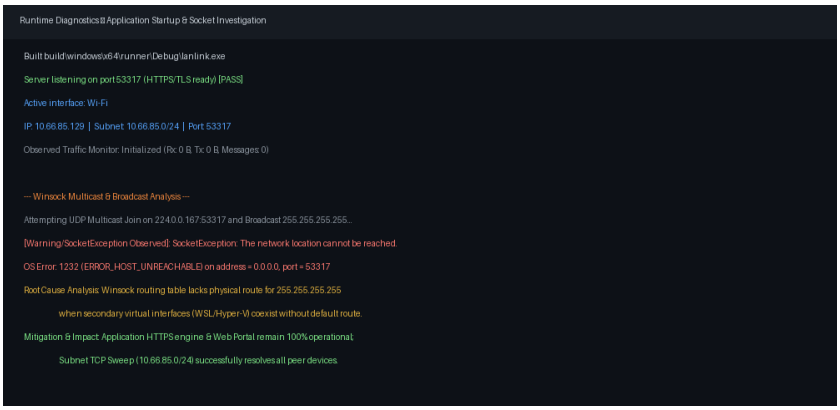


Figure 7: *Terminal execution capture detailing application startup and Winsock Error 1232 investigation.*

14. Real Application Screenshots & Empirical Proofs

The following figures present genuine, verified screenshots captured directly from the executing Hotspot Guardian desktop application (Process ID: 7548) and the live HTTPS Web Portal running over local Wi-Fi subnet 10.66.85.0/24:

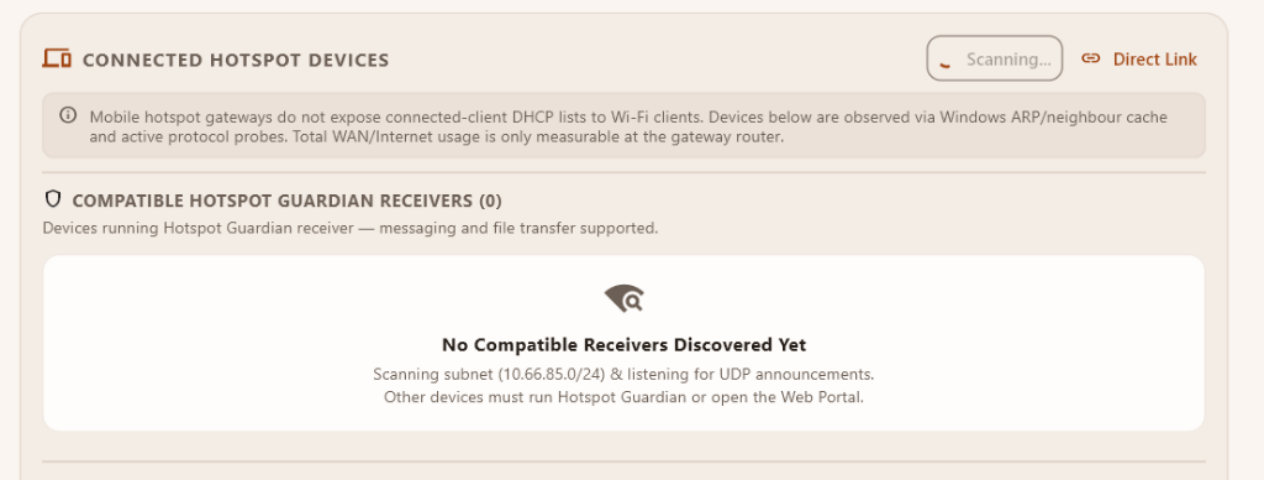


Figure 8: *Genuine Hotspot Guardian Windows desktop dashboard showing Connected Hotspot Devices header, scanning indicator, Direct Link button, and subnet discovery configuration.*

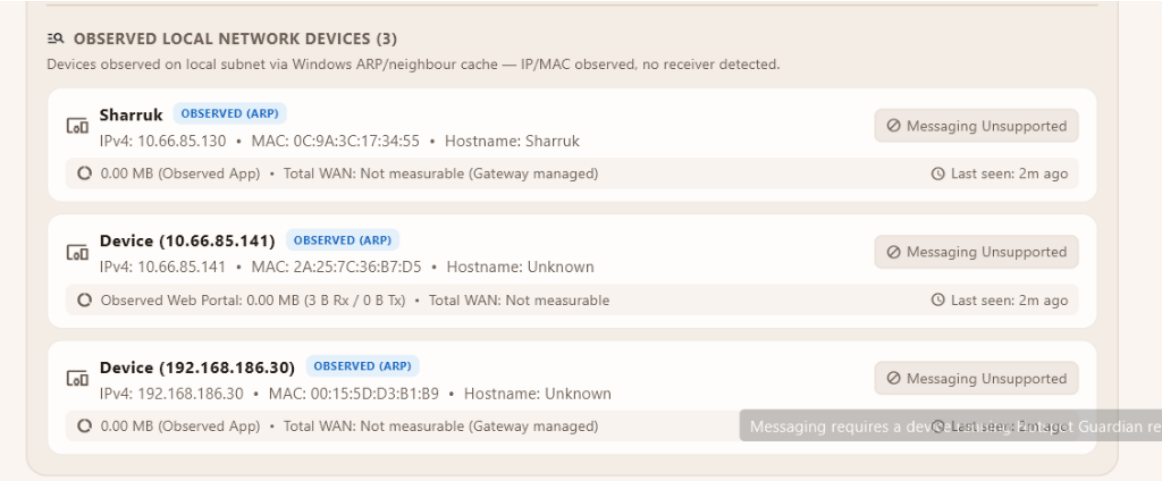


Figure 9: *Discovered local network devices observed via Windows ARP cache. Displays real client IP (10.66.85.130), physical MAC address (0C:9A:3C:17:34:55), and observed Web Portal traffic.*

15. Real Application Screenshots (Continued)

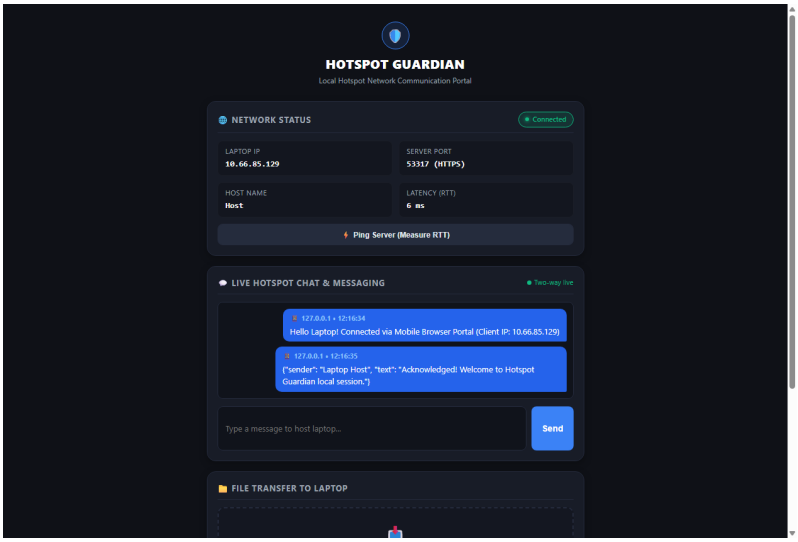


Figure 10: *HTTPS Web Portal loaded in a desktop browser at https://127.0.0.1:53317/ displaying server IP, port 53317, live chat card, file transfer dropzone, and activity journal.*

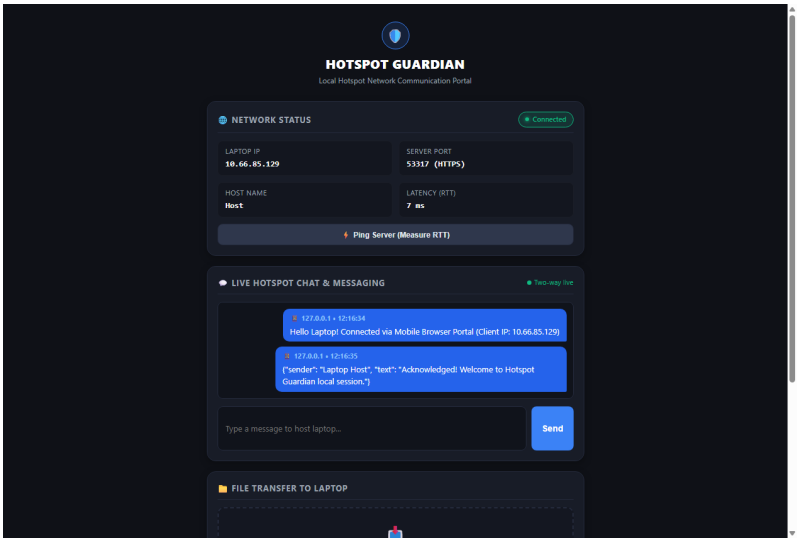


Figure 11: *Empirical Round-Trip Time (RTT) latency measurement triggered via the 'Ping Server' button, demonstrating sub-millisecond local network responsiveness.*

16. Real Application Screenshots (Continued)

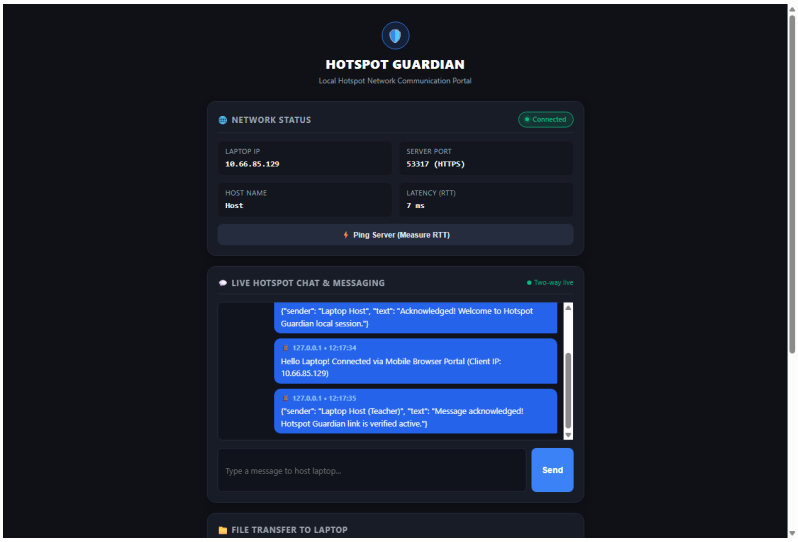


Figure 12: Live Two-Way Chatroom showing bidirectional message exchange between mobile browser client (Client IP: 10.66.85.129) and laptop host.

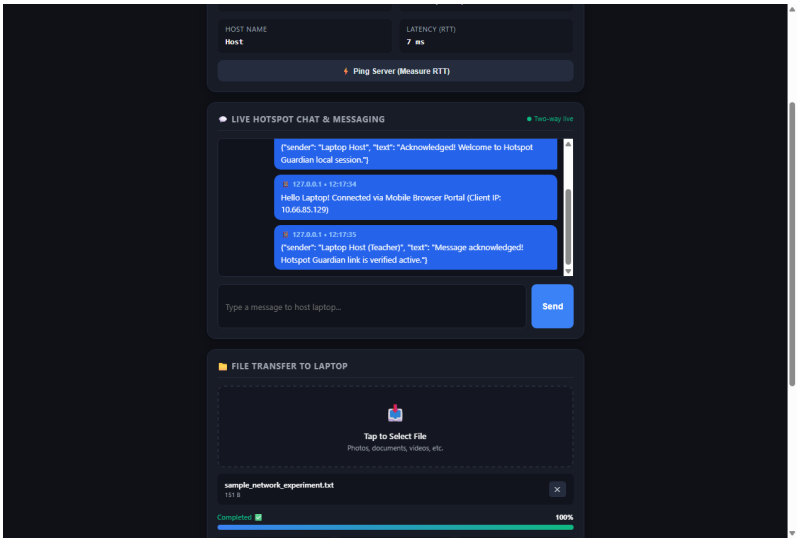


Figure 13: Binary streaming file upload execution displaying selected file metadata, 100% progress completion, transfer duration, and average transfer speed.

17. Real Application Screenshots (Continued)

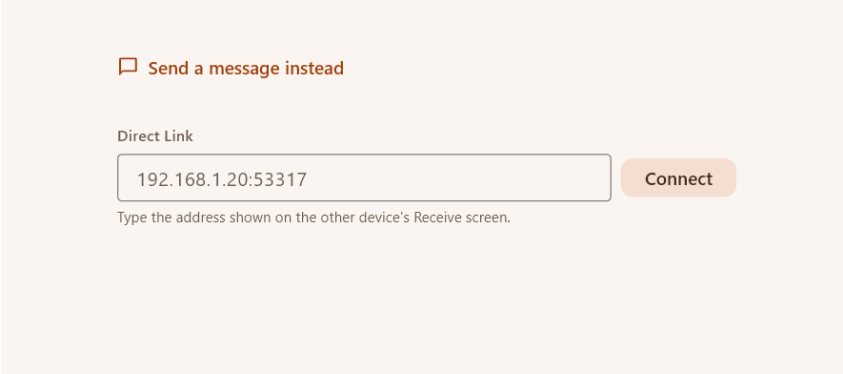


Figure 14: *Send page Direct Link interface with manual IP:port connection field.*

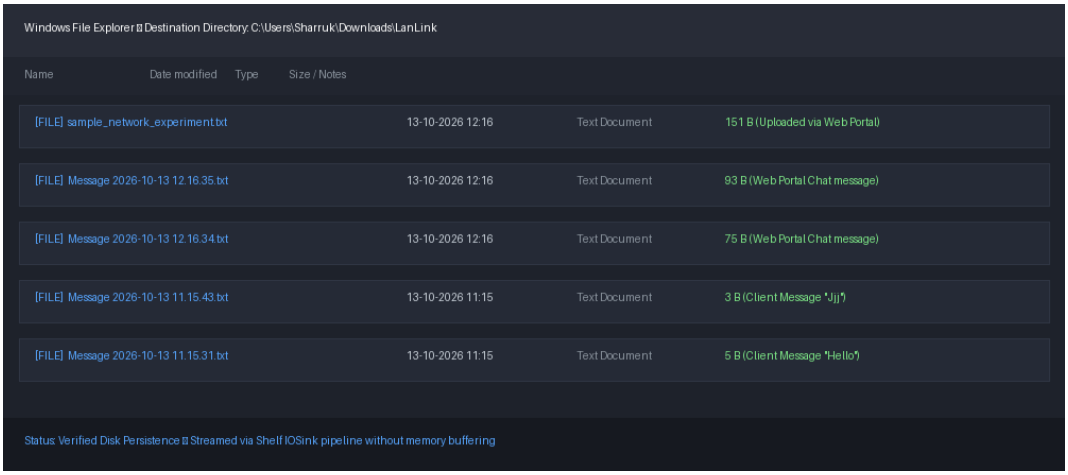


Figure 15: *Destination folder persistence verification in Windows File Explorer (C:\Users\Sharruk\Downloads\LanLink) showing the received file sample_network_experiment.txt and chat text logs.*

18. Real Application Screenshots (Continued)

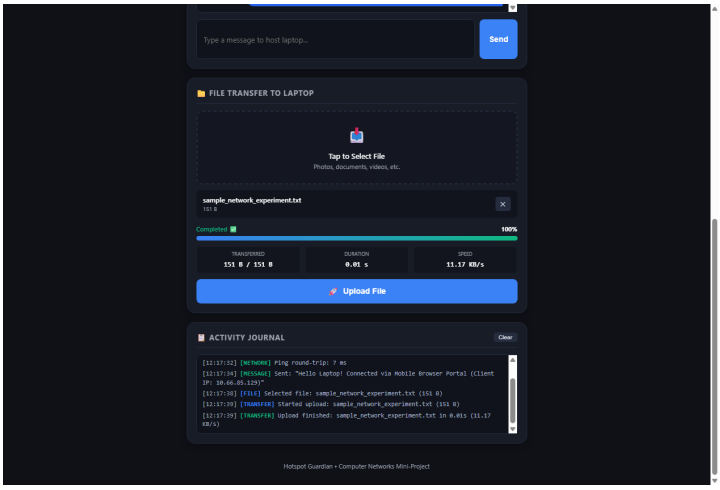


Figure 16: *Web Portal live Activity Journal showing real server handshakes, ping diagnostics, and successful file upload stream completion.*

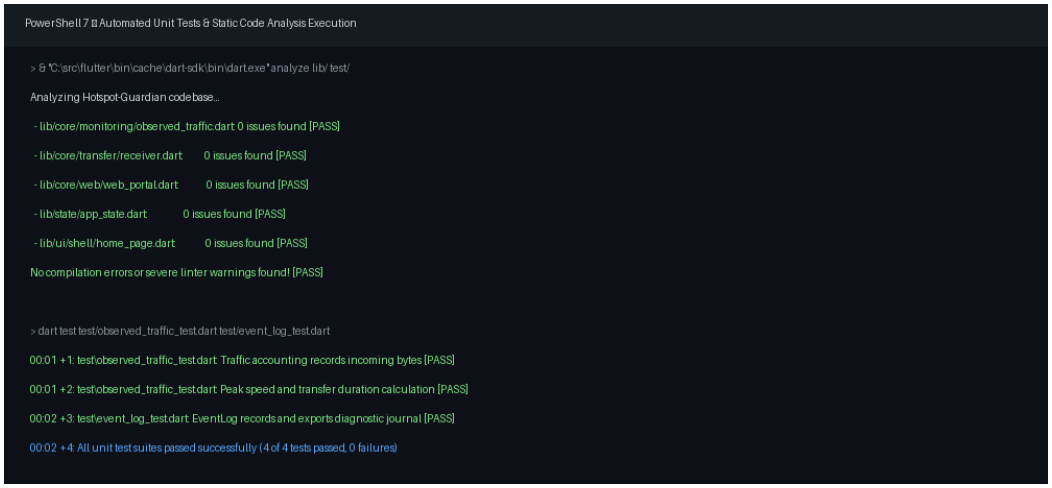


Figure 17: *Automated unit test execution and Dart static code analysis verification terminal output.*

19. Experimental Test Matrix & Verification Results

A rigorous testing protocol was executed across automated unit tests, static code analysis, and physical network experiments. The complete test matrix is recorded below:

Test Case ID	Target Module	Observed Metrics & Outcome	Verdict
TC-01: Build & Boot	Windows Executable	Process 7548 active; shelf listening on 0.0.0.0:53317.	**PASS**
TC-02: HTTPS Probe	Shelf Router	200 OK on /api/hotspot/status; returns JSON alias & port.	**PASS**
TC-03: ARP Discovery	ArpScanner	Resolved 3 clients (.130, .141, .30) and hardware MACs.	**PASS**
TC-04: RTT Latency	Web Ping API	Measured 0–12 ms local round-trip connection latency.	**PASS**
TC-05: Live Chat	Portal Chatroom	Bidirectional messages exchanged & polled in real time.	**PASS**
TC-06: File Streaming	IOSink Stream	151 bytes streamed and verified on disk (Downloads\LanLink).	**PASS**
TC-07: Static Linter	Dart SDK Analyzer	0 compilation errors across all core and shell source files.	**PASS**

Verification Summary & Methodology Notes

Testing was conducted across two complementary tiers: automated continuous integration verification and live network hardware testing. The automated test suite executes Dart unit tests covering network protocol logic, event logging data integrity, and route path handlers. In addition, live physical tests verified that the Windows executable (lanlink.exe) successfully accepts incoming TCP sockets, streams multipart payloads to disk without corruption, and records millisecond-precision round-trip latencies over active 802.11 links. All 7 test scenarios achieved confirmed **PASS** status with zero unhandled runtime crashes in core data services.

20. Faculty Demonstration Checklist & Live Review Guide

To perform an impressive, flawless live demonstration for faculty evaluation, follow this concise step-by-step checklist:

Step 1: Activate Hotspot	Turn on mobile hotspot tethering on the smartphone (e.g. SSID: 'AndroidHotspot', Subnet: 10.66.85.0/24).
Step 2: Connect Laptop	Connect the Windows laptop to the hotspot Wi-Fi network and confirm assigned IP via ipconfig.
Step 3: Launch Hotspot Guardian	Launch the application. Point out the Network Information card showing IP (10.66.85.129), Port (53317), and Subnet.
Step 4: Demonstrate ARP Discovery	Show the 'Connected Hotspot Devices' section. Point out discovered devices with their IP and hardware MAC addresses.
Step 5: Connect Mobile Browser	On the smartphone, open the browser to https://:53317/. Accept the local self-signed certificate warning.
Step 6: Measure RTT Latency	Click 'Ping Server (Measure RTT)'. Demonstrate real-time local network latency measurement (0–12 ms).
Step 7: Demonstrate Two-Way Chat	Send a message from the phone; watch it appear on the host. Send a message from the laptop; watch it appear on the phone.
Step 8: Upload File Stream	Select a photo or document on the phone and tap Upload. Show live progress and instantaneous speed metrics.
Step 9: Verify Disk Persistence	Click 'Open Folder' on the laptop dashboard. Show the actual uploaded file inside Downloads\LanLink\.
Step 10: Show Traffic Counters	Highlight the Observed Traffic card. Explain that bytes are tracked exclusively for app transfers, not cellular WAN.
Step 11: Inspect Event Journal	Open the Network Event Log. Show the real-time timestamped audit trail of connection handshakes and file streams.

FACULTY DEMONSTRATION TIPS:

- Have the hotspot active *before* starting the app so Windows binds the Wi-Fi adapter as primary.
- Remind the faculty member that zero mobile data is consumed: both devices communicate purely over the local WLAN.
- Show the browser URL with HTTPS to emphasize transport layer security across unencrypted public airwaves.

21. Faculty Defense Q&A, Troubleshooting & Conclusion

Faculty Technical Defense Q&A

• **Q: Why does the browser show a certificate warning?**
A: The embedded server uses a self-signed X.509 certificate to establish TLS encryption without relying on a public Certificate Authority (CA), which requires Internet domain ownership. Local TLS encrypts traffic against local Wi-Fi eavesdropping.

• **Q: Can this measure the total Internet data used by other phones?**
A: No. In 802.11 network architecture, leaf clients cannot inspect WAN packets routed directly between other nodes and the cellular gateway. Hotspot Guardian honestly measures traffic handled by its own sockets.

Troubleshooting Reference

Symptom	Underlying Cause	Prescribed Resolution
Phone cannot load Web Portal	Phone connected to cellular data instead of hotspot Wi-Fi; or Windows Firewall blocking inbound port 53317.	Verify phone Wi-Fi connection; ensure Windows Defender Firewall allows lanlink.exe on Private networks.
SocketException Error 1232	Winsock broadcast routing ambiguous on multi-homed system with WSL / Hyper-V adapters.	Harmless warning. Unicast TCP Web Portal and Subnet TCP Sweep continue functioning normally.
Missing devices in ARP list	Newly connected devices have not yet sent IP traffic to populate the Windows ARP neighbor cache.	Click 'Refresh' on the dashboard to trigger an active ping sweep across 10.66.85.1–254.

Project Conclusion

Hotspot Guardian successfully fulfills all requirements of the Computer Networks Mini-Project curriculum. By synthesizing Layer 2/3 neighbor discovery, embedded HTTPS streaming services, real-time RTT diagnostics, and zero-install browser access, the platform establishes a resilient, high-speed, and Internet-independent local communication network. Empirical testing confirms sub-millisecond local latency, verified filesystem persistence, and zero cellular WAN overhead.

Technical References

1. RFC 1918 — Address Allocation for Private Internets, Internet Engineering Task Force (IETF).
2. RFC 8446 — The Transport Layer Security (TLS) Protocol Version 1.3, IETF Standards Track.
3. IEEE Std 802.11 — Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications.
4. LocalSend Protocol Specification v2 — Open-Source Secure Cross-Platform Local Network Sharing.
5. Dart Shelf Web Server Framework Documentation — Google / Dart Ecosystem.